OM BHARADWAJ

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SUMMARY

Computer Science undergraduate at IIIT Manipur (CGPA 9.05) with hands-on experience in full-stack development and applied machine learning. I build models and then ship them as working products: Ocula (diabetic retinopathy screening, with a first-author paper under review at an IEEE conference) and Cutis (skin lesion screening on ISIC 2024). At Tata Technologies I built an enterprise CRM portal. Looking for software engineering, backend or applied AI roles.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, Java, SQL, PHP, C
Backend & Web: FastAPI, Node.js, Express.js, PHP, REST APIs, Microservices, JWT auth, role-based access control, React.js
Databases: PostgreSQL, MySQL, MongoDB, SQLite
Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, SVM, ADASYN, Pandas, NumPy
Models: EfficientNet-B4, ResNet-18, DenseNet-121, GoogLeNet, ConvNeXt, Swin-T, EdgeNeXt, EVA02
Explainability & LLMs: Grad-CAM, LIME, SHAP, Groq and Gemini APIs
DevOps: Docker, Docker Compose, Git, GitHub, CI/CD, Render
Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

PROJECTS

Ocula: Diabetic Retinopathy Screening Platform GitHub
Python, PyTorch, FastAPI, Node.js/Express, React, PostgreSQL, Docker

- Built a full-stack screening platform. On APTOS 2019 it reaches 98.50% binary accuracy (AUC 0.997) and 90.68% five-class accuracy, which is 0.70 and 1.39 points above the published Butt et al. (2022) baseline. The work is the basis of my first-author paper under review at an IEEE conference.
- Ensembled five CNN backbones (ResNet-18, GoogLeNet, EfficientNet-B4, DenseNet-121, ConvNeXt-Tiny) with 8-view test-time augmentation and soft-voting classical classifiers (SVM, RF, XGBoost). Binary accuracy rose 1.38 points over the best single backbone.
- Designed a two-level hierarchical SVM cascade with ADASYN oversampling for five-class grading. It gained 1.27 points over a flat classifier and lifted recall on the minority Severe DR class from 0.79 to 0.82.
- Added Grad-CAM, LIME and SHAP explanations plus two Groq LLaMA-3 reports (clinical and patient); a full prediction with explanations takes under 45 seconds on CPU.
- Deployed as four Docker Compose services (React, Node.js/Express, FastAPI, PostgreSQL) with JWT auth and role-based access, on a validated clinical dataset of 3,662 images.

Cutis: Skin Lesion Screening Platform GitHub
Python, PyTorch, LightGBM, FastAPI, Node.js/Express, React, SQLite, Docker

- Built a skin-cancer risk pipeline on ISIC 2024 (401,059 lesions, 393 malignant, 1,042 patients). With patient-grouped 5-fold validation it reaches AUROC 0.958 and pAUC (≥80% TPR) 0.165. A late Kaggle submission scored 0.174 public and 0.160 private.
- Designed a two-stage cascade. A tabular GBDT gate (LightGBM, XGBoost, CatBoost on 153 features, including patient-context features) clears 50% of lesions at 98.7% malignant recall. The rest go to a 4-backbone CNN ensemble (ConvNeXt-Tiny, Swin-T, EdgeNeXt, EVA02; 5 folds each) fused by a stage-2 LightGBM.
- Ran a leakage audit showing how image-level splits inflate results (97.6% vs 64.0% patient-grouped nearest-neighbour accuracy). Used paired bootstrap CIs and leave-one-institution-out validation, and reported the honest numbers instead of augment-then-split ones.
- Deployed as four Docker Compose services (React, Node.js/Express, FastAPI, SQLite) with role-based access for patients and clinicians. Each scan gets Grad-CAM, LIME and SHAP explanations in parallel, a role-specific Groq LLM report, scan history with trend and a PDF download.
- Added safeguards: patients never see raw scores or cancer types, the LLM never receives the photo or patient ID, and urgency text follows the risk tier with a template fallback. Covered by 51 automated tests.

EXPERIENCE

Tata Technologies May 2026 to July 2026
Software Engineering Intern *Jamshedpur, Jharkhand, India*

- Worked in an agile team on an enterprise CRM portal that large clients use to manage their customer relationships.
- Built a responsive React.js frontend with dashboards, charts and real-time notifications.
- Wrote REST APIs in PHP with role-based access control, authentication middleware and automated reporting.
- Designed MySQL schemas with normalized tables, indexed queries and stored procedures so the system could scale.

EDUCATION

Indian Institute of Information Technology Manipur Imphal, India
B.Tech, Computer Science & Engineering (AI & Data Science) *2023 to 2027 (Expected)*
CGPA: 9.05 / 10.0 (Top 5% of the cohort)

ACHIEVEMENTS & LEADERSHIP

- Top 5% of my cohort at IIIT Manipur across five semesters, with a CGPA of 9.05.
- GDG On Campus Chapter Lead, IIIT Manipur. Selected through Google's competitive process. I run Study Jams, workshops and developer events for 1200+ students.
- GeeksforGeeks Campus Ambassador. I connect students with technical resources and competitive programming initiatives, and I solve problems regularly on LeetCode.